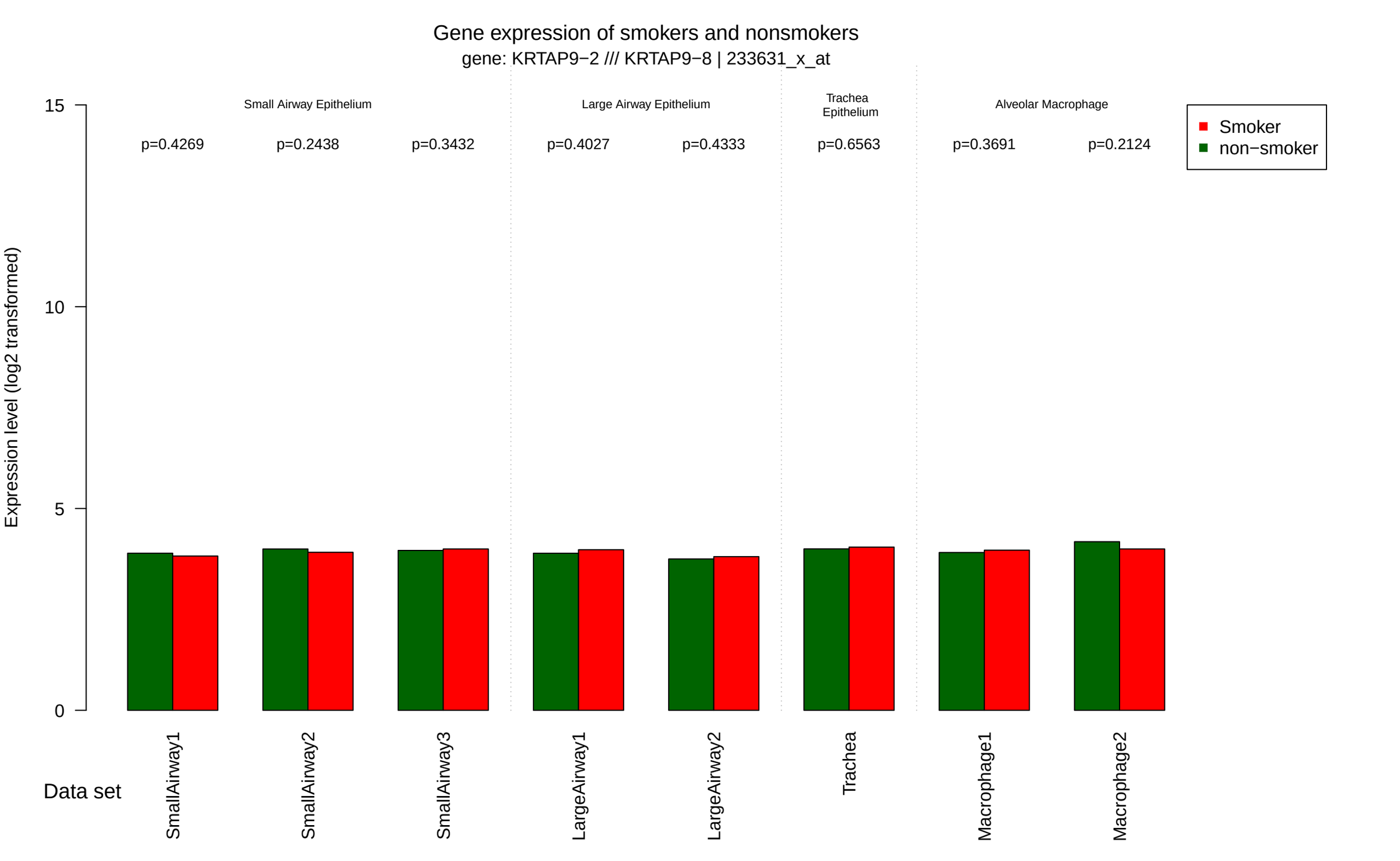
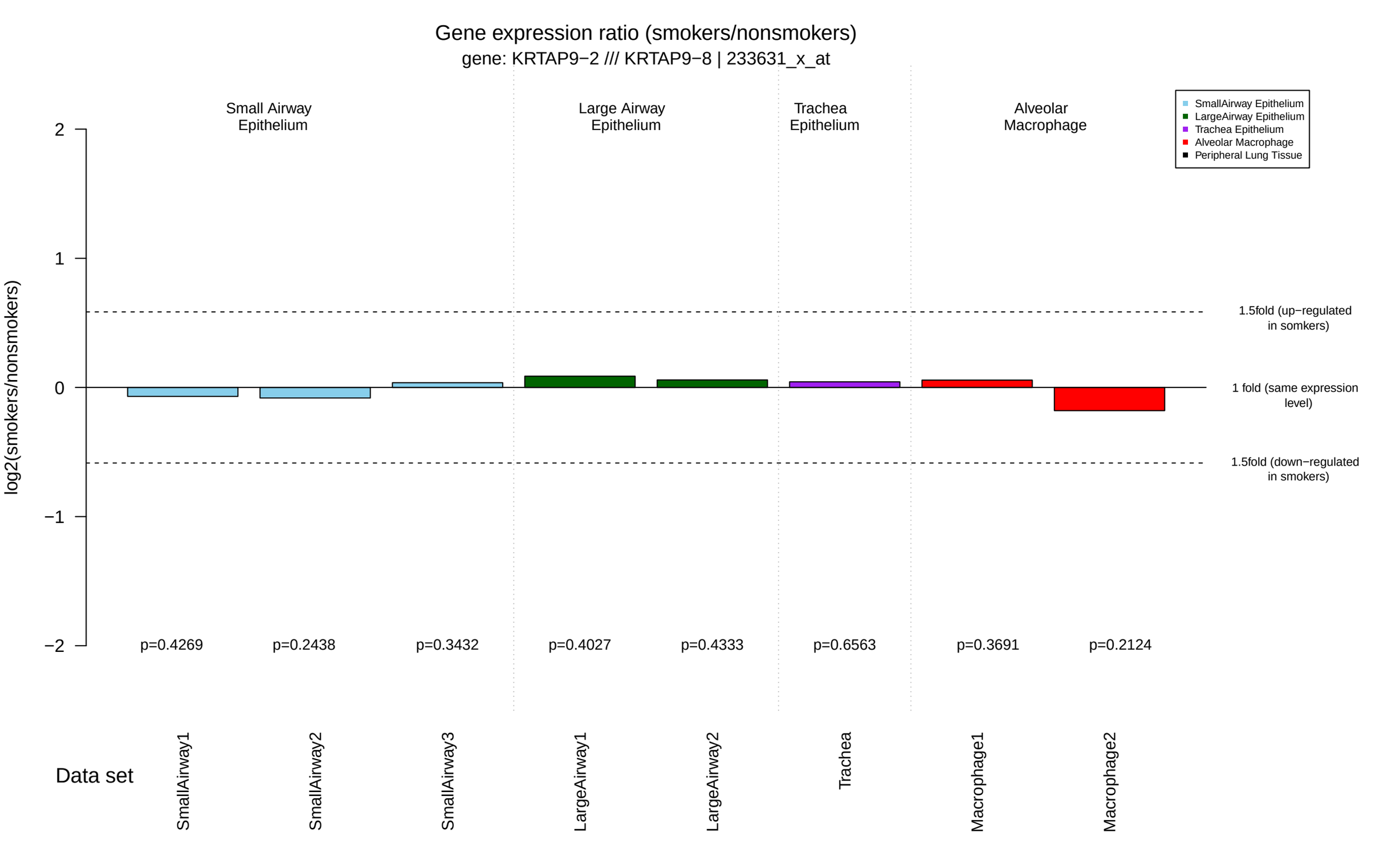
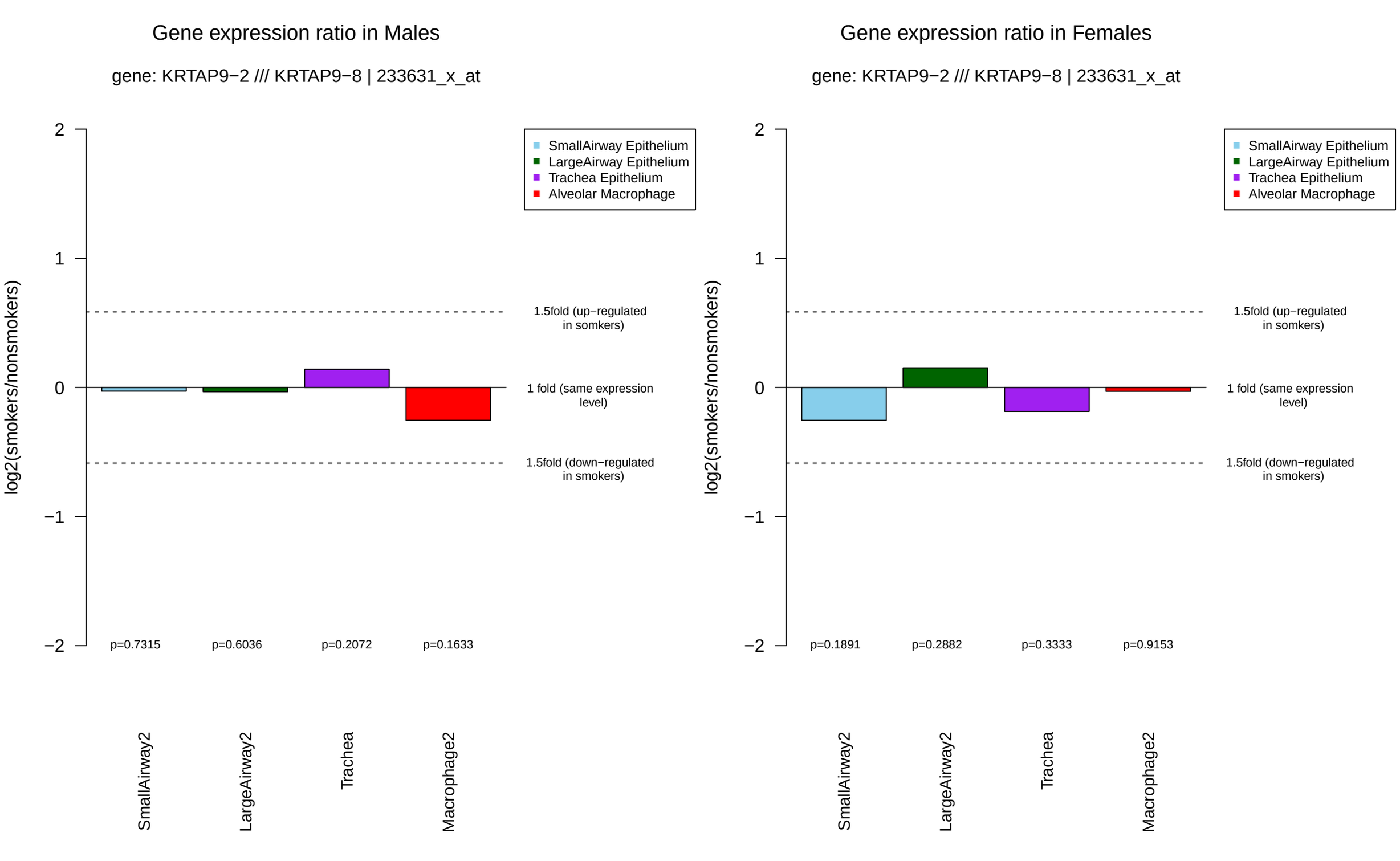
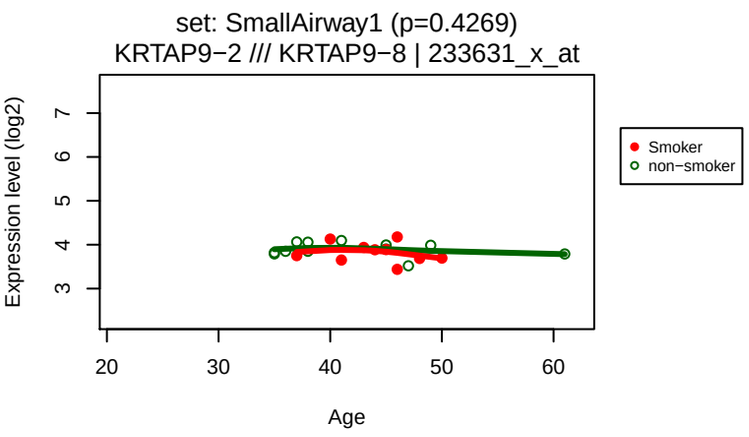
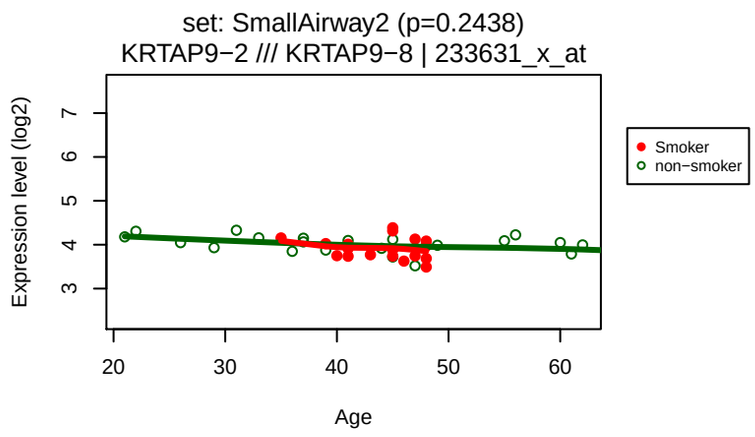
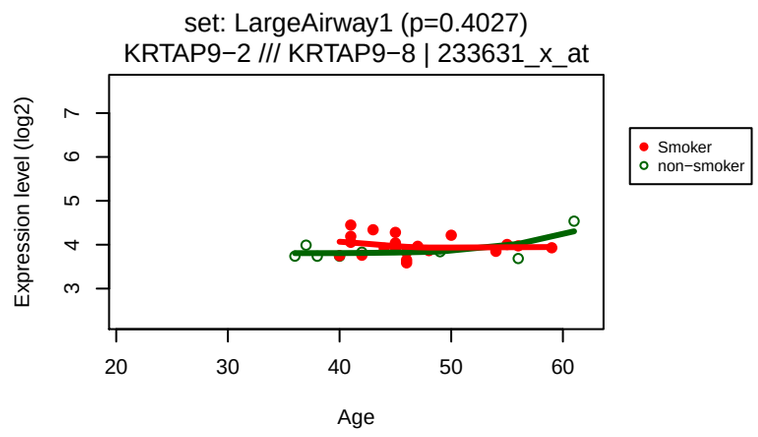

Gene expression of smokers and nonsmokers
gene: KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4269
p=0.2438
p=0.3432
p=0.4027
p=0.4333
p=0.6563
p=0.3691
p=0.2124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

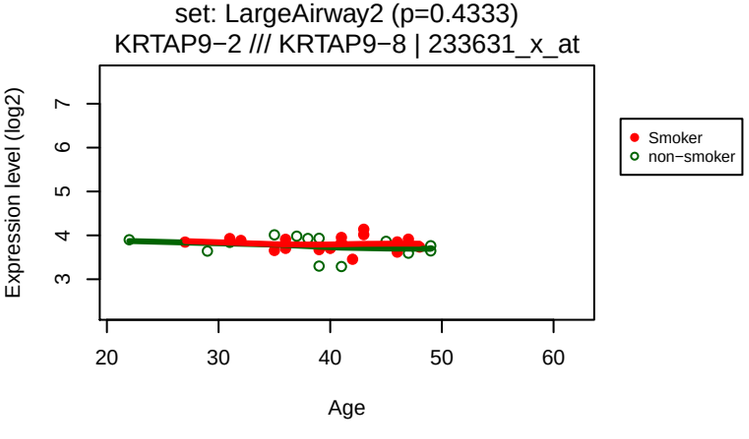

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4269
p=0.2438
p=0.3432
p=0.4027
p=0.4333
p=0.6563
p=0.3691
p=0.2124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

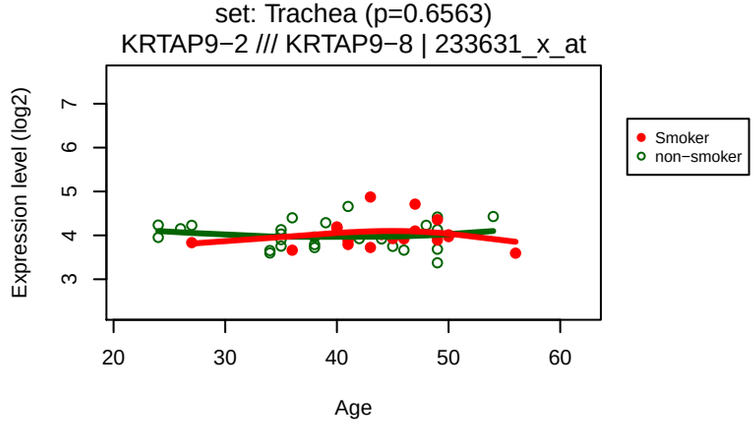

Gene expression ratio in Males
gene: KRTAP9−2 /// KRTAP9−8 | 233631_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7315
p=0.6036
p=0.2072
p=0.1633
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP9−2 /// KRTAP9−8 | 233631_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1891
p=0.2882
p=0.3333
p=0.9153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

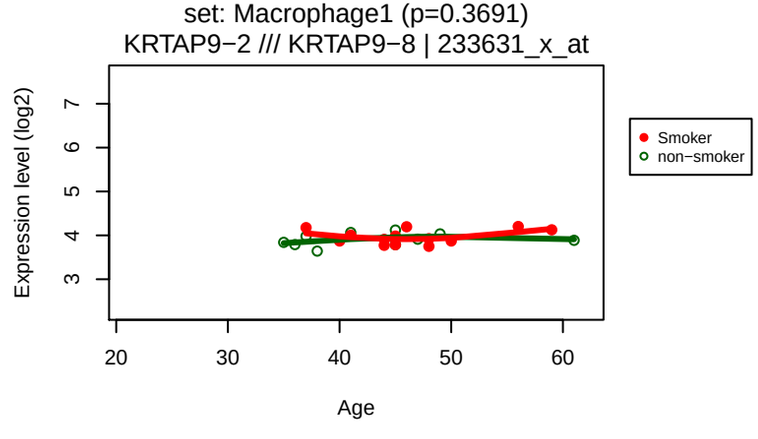

set: SmallAirway1 (p=0.4269)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker

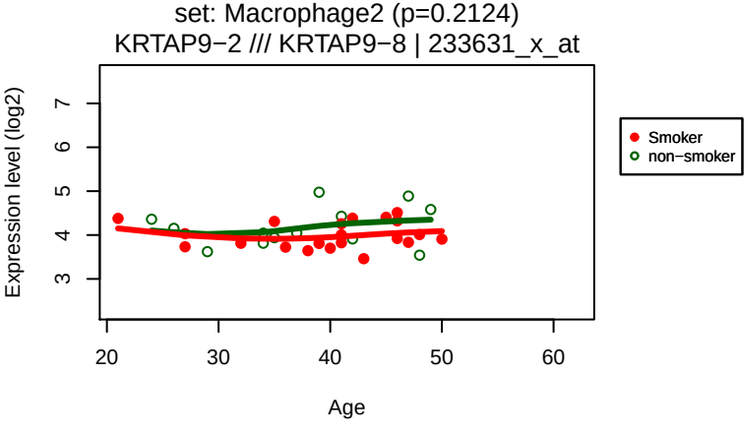

set: SmallAirway2 (p=0.2438)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker

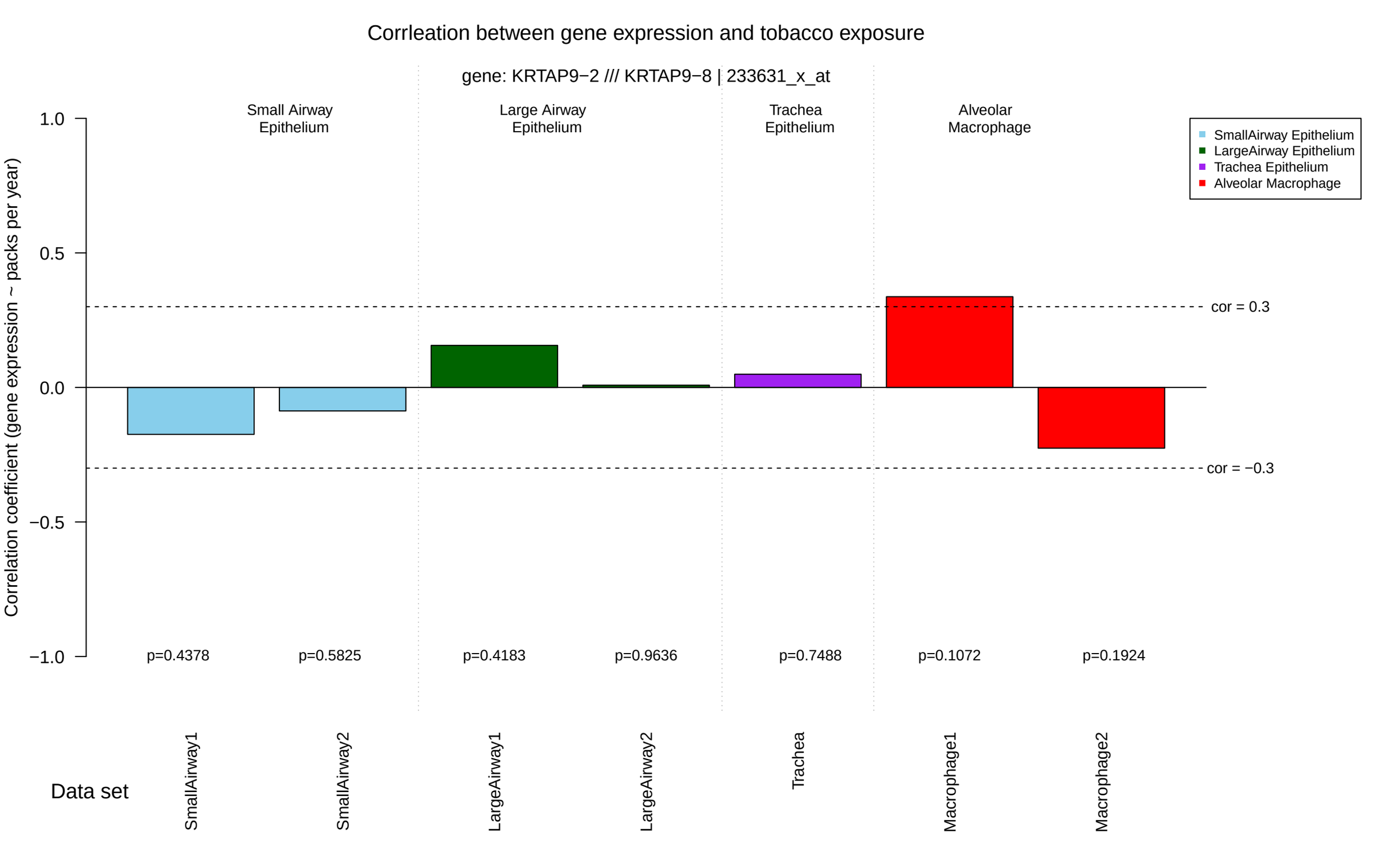

set: LargeAirway1 (p=0.4027)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4333)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6563)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3691)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2124)
KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP9−2 /// KRTAP9−8 | 233631_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4378
p=0.5825
p=0.4183
p=0.9636
p=0.7488
p=0.1072
p=0.1924
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2